

SCIENCE ALL SETS 2026

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CHAPTER: 1 - Life Processes (Biology)

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YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which structure in a leaf is mainly responsible for gaseous exchange?

OPTIONS: (A) Xylem (B) Stomata (C) Phloem (D) Cuticle

ANSWER: (B) Stomata

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a correct statement about Spirogyra, Leishmania and Hydra?

OPTIONS: (A) They reproduce sexually (B) They are unicellular organisms (C) They are multicellular organisms (D) They reproduce asexually

ANSWER: (D) They reproduce asexually

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which part of the human brain is responsible for the precision of voluntary actions?

OPTIONS: (A) Cerebrum (B) Cerebellum (C) Medulla (D) Pons

ANSWER: (B) Cerebellum

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Pancreas secretes pancreatic juice containing enzymes. Which option correctly describes their functions?

OPTIONS: (A) Trypsin digests emulsified fats; Lipase digests proteins (B) Trypsin digests proteins; Lipase digests emulsified fats (C) Both Trypsin and Lipase digest fats (D) Trypsin digests proteins; Lipase digests carbohydrates

ANSWER: (B) Trypsin digests proteins; Lipase digests emulsified fats

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ORIGINAL_MARKS: 1

QUESTION: In human beings, sex is determined genetically. Which combination is correct for determining the sex of a child?

OPTIONS: (A) XX from mother and XX from father gives male child (B) XX from mother and XY from father gives female child (C) XX from mother and X from father gives female child; XX from mother and Y from father gives male child (D) XY from mother and XX from father gives male child

ANSWER: (C) XX from mother and X from father gives female child; XX from mother and Y from father gives male child

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following groups contains only non-biodegradable materials?

OPTIONS: (A) Vegetable peels, dead leaves, paper (B) Cow dung, leather bag, water (C) Polythene bag, rubber band, ball pen (D) Paper, fruits, bones

ANSWER: (C) Polythene bag, rubber band, ball pen

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Reflex actions do not involve thinking. Reason (R): Most reflex actions are controlled by the spinal cord.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): In human beings, the respiratory pigment is haemoglobin present in red blood cells. Reason (R): Haemoglobin has a very high affinity for carbon dioxide.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (C) A is true but R is false

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the function of the diaphragm in the human respiratory system and where is it located?

OPTIONS: (A) It helps in gaseous exchange between blood and tissues; located in the chest cavity (B) It contracts and relaxes to change thoracic cavity volume during breathing; located below the lungs (C) It filters air entering the respiratory tract; located in the nasal cavity (D) It produces sound during speech; located in the larynx

ANSWER: (B) It contracts and relaxes to change thoracic cavity volume during breathing; located below the lungs

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which statement correctly differentiates between chewing food and salivation on sight of food?

OPTIONS: (A) Chewing is voluntary while salivation is involuntary (B) Chewing is involuntary while salivation is voluntary (C) Both are voluntary actions (D) Both are involuntary actions

ANSWER: (A) Chewing is voluntary while salivation is involuntary

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which statement correctly differentiates between pollination and fertilization?

OPTIONS: (A) Pollination is fusion of gametes; fertilization is transfer of pollen grains (B) Pollination is transfer of pollen grains; fertilization is fusion of gametes (C) Both are the same process (D) Pollination occurs after fertilization

ANSWER: (B) Pollination is transfer of pollen grains; fertilization is fusion of gametes

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Mendel crossed tall pea plants with short pea plants and obtained all tall plants in F1 generation. Why were only tall plants observed?

OPTIONS: (A) Tall trait is dominant over short trait (B) Short trait is dominant over tall trait (C) Both traits are equally dominant (D) The tall trait disappeared in F1

ANSWER: (A) Tall trait is dominant over short trait

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: How did Mendel obtain F2 progeny from the F1 generation?

OPTIONS: (A) By cross-pollinating F1 plants with dominant trait plants (B) By cross-pollinating F1 plants with recessive trait plants (C) By self-pollinating F1 generation plants (D) By cross-pollinating both parent plants

ANSWER: (C) By self-pollinating F1 generation plants

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which of the following correctly describes the difference between dominant and recessive traits?

OPTIONS: (A) Dominant trait is expressed only in homozygous condition; recessive trait is expressed in both homozygous and heterozygous conditions (B) Dominant trait is expressed in both homozygous and heterozygous conditions; recessive trait is expressed only in homozygous condition (C) Both traits are expressed only in homozygous condition (D) Both traits are expressed in heterozygous condition

ANSWER: (B) Dominant trait is expressed in both homozygous and heterozygous conditions; recessive trait is expressed only in homozygous condition

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: If a person is unable to produce insulin, what is the likely impact on their body?

OPTIONS: (A) Blood sugar level decreases dangerously (B) Blood sugar level increases leading to diabetes (C) Blood pressure increases (D) Digestion of proteins is affected

ANSWER: (B) Blood sugar level increases leading to diabetes

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: If the kidney fails to filter blood properly, what would happen in the body?

OPTIONS: (A) Oxygen transport would be affected (B) Nitrogenous wastes would accumulate in the blood (C) Digestion of food would stop (D) Blood pressure would decrease

ANSWER: (B) Nitrogenous wastes would accumulate in the blood

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The genotype of green stemmed tomato plant is 'GG' and purple stemmed plant is 'gg'. When F1 hybrids are self-pollinated, what is the ratio of GG, Gg and gg in F2 progeny?

OPTIONS: (A) 2:1:1 (B) 3:1:0 (C) 1:1:2 (D) 1:2:1

ANSWER: (D) 1:2:1

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The movement of pollen tube towards an ovule is an example of:

OPTIONS: (A) Hydrotropism (B) Geotropism (C) Chemotropism (D) Phototropism

ANSWER: (C) Chemotropism

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: How does Spirogyra reproduce asexually?

OPTIONS: (A) By division of cells into two cells (B) By division of cells into many cells
(C) By breaking up of filaments into smaller parts (D) By formation of gametes after
meiosis

ANSWER: (C) By breaking up of filaments into smaller parts

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Carbohydrates not used immediately in plants are stored as:

OPTIONS: (A) Starch (B) Sugar (C) Fat (D) Protein

ANSWER: (A) Starch

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which events occur during photosynthesis? (i) Oxidation of CO₂ to
carbohydrates (ii) Conversion of light energy to chemical energy (iii) Absorption of light
energy by chlorophyll (iv) Conversion of H₂ and O₂ to water

OPTIONS: (A) (i) and (ii) (B) (ii) and (iii) (C) (iii) and (iv) (D) (i) and (iv)

ANSWER: (B) (ii) and (iii)

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which factor limits the number of trophic levels in a food chain?

OPTIONS: (A) Availability of water (B) Deficient food supply (C) Decrease in energy at
higher levels (D) Pollution in air

ANSWER: (C) Decrease in energy at higher levels

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Blood plasma transports CO_2 in dissolved form while O_2 is transported by respiratory pigments. Reason (R): CO_2 is more soluble in water than O_2 .

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): There are generally more individuals at lower trophic levels. Reason (R): The greatest number is of the producers.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (B) Both A and R are true but R is not the correct explanation of A

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ORIGINAL_MARKS: 1

QUESTION: Tomato is a natural source of which acid?

OPTIONS: (A) Acetic acid (B) Tartaric acid (C) Oxalic acid (D) Citric acid

ANSWER: (B) Tartaric acid

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Rain water is called acid rain when its pH:

OPTIONS: (A) falls below 7.2 (B) falls below 14 (C) is above 7 (D) falls below 5.6

ANSWER: (D) falls below 5.6

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ORIGINAL_MARKS: 1

QUESTION: Which of the following are combination reactions? (i) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$

(ii) $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$ (iii) $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ (iv) $\text{Zn} + \text{FeSO}_4 \rightarrow \text{ZnSO}_4 + \text{Fe}$

OPTIONS: (A) (i) and (iii) (B) (iii) and (iv) (C) (ii) and (iv) (D) (ii) and (iii)

ANSWER: (D) (ii) and (iii)

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A metal carbonate reacts with solution X to form salt, water and gas Y. X and Y are:

OPTIONS: (A) X-Sodium hydroxide, Y-Carbon dioxide (B) X-Sodium hydroxide, Y-Hydrogen (C) X-Hydrochloric acid, Y-Carbon dioxide (D) X-Hydrochloric acid, Y-Hydrogen

ANSWER: (C) X-Hydrochloric acid, Y-Carbon dioxide

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When ferrous sulphate crystals are heated in a test tube, what is observed?

OPTIONS: (A) A colourless gas with no smell is evolved (B) A brown gas is evolved (C) The green colour fades and a gas with smell of burning sulphur is evolved (D) The green colour fades and no gas is evolved

ANSWER: (C) The green colour fades and a gas with smell of burning sulphur is evolved

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When Mg metal is treated with hot water, it starts floating because:

OPTIONS: (A) Mg is lighter than water (B) Mg reacts with water to produce H_2 gas which helps in floating (C) Mg reacts with water to produce N_2 gas which helps in floating (D) Mg reacts with water to form CO_2 gas which helps in floating

ANSWER: (B) Mg reacts with water to produce H_2 gas which helps in floating

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The pH of gastric juices released during digestion is:

OPTIONS: (A) Less than 7 (B) More than 7 (C) Equal to 7 (D) Equal to 0

ANSWER: (A) Less than 7

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Baking powder is a mixture of:

OPTIONS: (A) Sodium carbonate + Acetic acid (B) Sodium carbonate + Tartaric acid (C) Sodium hydrogen carbonate + Tartaric acid (D) Sodium hydrogen carbonate + Acetic acid

ANSWER: (C) Sodium hydrogen carbonate + Tartaric acid

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Silver jewellery turns dull with a grey-black coating after a few months due to:

OPTIONS: (A) Removal of polish (B) Corrosion due to air, moisture and acids (C) Dust particles (D) Rusting

ANSWER: (B) Corrosion due to air, moisture and acids

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Carbon compounds can form chains, branched and ring structures. Reason (R): Carbon exhibits the property of catenation.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Iron filings were added to four test tubes containing ZnSO_4 , CuSO_4 , FeSO_4 and $\text{Al}_2(\text{SO}_4)_3$. In which test tube would the reaction be most vigorous?

OPTIONS: (A) ZnSO_4 (B) CuSO_4 (C) FeSO_4 (D) $\text{Al}_2(\text{SO}_4)_3$

ANSWER: (B) CuSO_4

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the balanced equation for the reaction between iron and copper sulphate?

OPTIONS: (A) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ (B) $2\text{Fe} + 3\text{CuSO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{Cu}$ (C) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{Cu}$ (D) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{CuSO}_4$

ANSWER: (A) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which metal is so soft that it can be cut with a knife?

OPTIONS: (A) Iron (B) Copper (C) Sodium (D) Aluminium

ANSWER: (C) Sodium

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which non-metal is a good conductor of electricity?

OPTIONS: (A) Sulphur (B) Phosphorus (C) Graphite (D) Silicon

ANSWER: (C) Graphite

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Using electronic configuration, how does Mg combine with oxygen to form MgO?

OPTIONS: (A) Mg loses 2 electrons to O which gains 2 electrons (B) Mg gains 2 electrons from O which loses 2 electrons (C) Both share electrons equally (D) Mg loses 1 electron and O gains 1 electron

ANSWER: (A) Mg loses 2 electrons to O which gains 2 electrons

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Why should metal sulphides and carbonates be converted to metal oxides before extraction?

OPTIONS: (A) Oxides are easier to reduce than sulphides and carbonates (B) Oxides are more abundant in nature (C) Sulphides and carbonates are non-reactive (D) Oxides have higher melting points

ANSWER: (A) Oxides are easier to reduce than sulphides and carbonates

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Why is aluminium oxide considered an amphoteric oxide?

OPTIONS: (A) It reacts only with acids (B) It reacts only with bases (C) It reacts with both acids and bases to form salt and water (D) It does not react with any substance

ANSWER: (C) It reacts with both acids and bases to form salt and water

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Why are metals like Na, K, Ca and Mg never found in free state in nature?

OPTIONS: (A) They are highly reactive and readily combine with other elements (B) They are insoluble in water (C) They have high melting points (D) They are radioactive

ANSWER: (A) They are highly reactive and readily combine with other elements

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: What happens when copper powder is heated in a china dish?

OPTIONS: (A) It melts to form liquid copper (B) It reacts with oxygen to form black copper oxide (C) It sublimes into gas (D) It remains unchanged

ANSWER: (B) It reacts with oxygen to form black copper oxide

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: When H_2 gas is passed over heated copper oxide, what is observed?

OPTIONS: (A) Black copper oxide turns to brown copper (B) Black copper oxide turns to white copper (C) No change is observed (D) Copper oxide melts

ANSWER: (A) Black copper oxide turns to brown copper

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: In electrolysis of brine, gas 'Y' burns with a pop sound. What is gas 'Y'?

OPTIONS: (A) Oxygen (B) Chlorine (C) Hydrogen (D) Nitrogen

ANSWER: (C) Hydrogen

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: In electrolysis of brine, gas 'X' is used for disinfecting drinking water. What is gas 'X'?

OPTIONS: (A) Hydrogen (B) Chlorine (C) Oxygen (D) Carbon dioxide

ANSWER: (B) Chlorine

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: A carbon compound P (C_2H_6O) is neutral to litmus and produces a gas with pop sound when reacted with sodium. Identify P.

OPTIONS: (A) Methanol (B) Ethanol (C) Ethanoic acid (D) Methanoic acid

ANSWER: (B) Ethanol

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: When ethanol is heated with conc. H_2SO_4 at 443 K, what is the role of H_2SO_4 ?

OPTIONS: (A) Oxidizing agent (B) Reducing agent (C) Dehydrating agent (D) Catalyst

ANSWER: (C) Dehydrating agent

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: What is observed when ethanol is heated with ethanoic acid in presence of an acid catalyst?

OPTIONS: (A) A sweet-smelling ester is formed (B) A foul-smelling gas is evolved (C) A precipitate is formed (D) No reaction occurs

ANSWER: (A) A sweet-smelling ester is formed

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Why can carbon neither form C^{4+} cation nor C^{4-} anion?

OPTIONS: (A) Carbon has 4 valence electrons and requires too much energy to lose or gain 4 electrons (B) Carbon is not reactive (C) Carbon is a non-metal (D) Carbon has only 2 shells

ANSWER: (A) Carbon has 4 valence electrons and requires too much energy to lose or gain 4 electrons

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: What type of bond is formed between carbon and hydrogen?

OPTIONS: (A) Ionic bond (B) Covalent bond (C) Metallic bond (D) Hydrogen bond

ANSWER: (B) Covalent bond

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which characteristics are associated with anaerobic respiration in human muscles? (i) Muscle cell (ii) Requires O_2 (iii) Produces 3-carbon molecules

OPTIONS: (A) (i) and (ii) (B) (i) and (iii) (C) (ii) and (iii) (D) (i) only

ANSWER: (B) (i) and (iii)

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Walking in a straight line and riding a bicycle are possible due to coordination of which brain part?

OPTIONS: (A) Fore-brain Cerebrum (B) Mid-brain Cerebellum (C) Hind-brain Cerebellum (D) Hind-brain Medulla

ANSWER: (C) Hind-brain Cerebellum

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a synapse, the chemical signal is transmitted from:

OPTIONS: (A) Dendritic end of one neuron to axonal end of another neuron (B) Axon to cell body of the same neuron (C) Cell body to axonal end of the same neuron (D) Axonal end of one neuron to dendritic end of another neuron

ANSWER: (D) Axonal end of one neuron to dendritic end of another neuron

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Organisms of a higher trophic level feeding on several types of organisms from a lower trophic level form the:

OPTIONS: (A) Food web (B) Ecosystem (C) Ecological pyramid (D) Food chain

ANSWER: (A) Food web

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In accumulation of toxic substances through a food chain, which organism is affected the most?

OPTIONS: (A) Small aquatic plants (B) Human beings (C) Small fish (D) Large fish

ANSWER: (B) Human beings

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The growth of tendrils in grape plants is due to:

OPTIONS: (A) Effect of light (B) Effect of gravity (C) Rapid cell division in cells in contact with support (D) Rapid cell division in cells away from support
ANSWER: (D) Rapid cell division in cells away from support

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CHAPTER: 1 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Natural replenishment of soil will not occur in absence of microbes. Reason (R): Microbes break down dead remains and waste products into the soil.
OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true
ANSWER: (A) Both A and R are true and R is the correct explanation of A

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CHAPTER: 1 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The basic filtration unit in kidneys is a cluster of thin-walled blood capillaries. Reason (R): Each capillary cluster in the kidney is associated with Bowman's capsule.
OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true
ANSWER: (B) Both A and R are true but R is not the correct explanation of A

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CHAPTER: 1 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In plants, the role of cytokinin is to:

OPTIONS: (A) promote cell division (B) cause wilting of leaves (C) promote opening of stomatal pore (D) promote growth of stem

ANSWER: (A) promote cell division

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of chromosomes in parents and offspring remains constant in sexual reproduction due to:

OPTIONS: (A) doubling of chromosomes after zygote formation (B) halving of chromosomes after zygote formation (C) doubling of chromosomes before gamete formation (D) halving of chromosomes at gamete formation

ANSWER: (D) halving of chromosomes at gamete formation

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ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The timing and amount of hormone secretion are regulated in the human body. Which example best illustrates this?

OPTIONS: (A) Insulin secretion increases when blood sugar rises and decreases when it falls (B) Adrenaline is secreted only during sleep (C) Growth hormone is secreted only in childhood (D) Thyroxine is secreted in constant amounts

ANSWER: (A) Insulin secretion increases when blood sugar rises and decreases when it falls

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CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What are two limitations of using electrical impulses for information transmission?

OPTIONS: (A) They travel too fast and cannot be controlled (B) They cannot reach all body parts and get fatigued easily (C) They can only travel in one direction and get weaker with distance (D) They require a medium and are slow

ANSWER: (C) They can only travel in one direction and get weaker with distance

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Variation is useful for species survival. Which example best explains this?

OPTIONS: (A) All individuals in a population are identical so they survive together (B) Different traits allow some individuals to survive environmental changes (C) Variation causes all individuals to die in changing environments (D) Variation is only useful for reproduction

ANSWER: (B) Different traits allow some individuals to survive environmental changes

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the best way to dispose of kitchen waste?

OPTIONS: (A) Burning in open air (B) Dumping in water bodies (C) Composting / Vermicomposting (D) Burying in soil

ANSWER: (C) Composting / Vermicomposting

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the best way to dispose of metallic waste?

OPTIONS: (A) Burning (B) Recycling (C) Dumping in landfills (D) Burying

ANSWER: (B) Recycling

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the best way to dispose of plastic bottles?

OPTIONS: (A) Burning (B) Recycling (C) Dumping (D) Composting

ANSWER: (B) Recycling

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Give two examples to justify that sex of offspring is not always genetically determined.

OPTIONS: (A) In humans and dogs, sex is temperature-dependent (B) In reptiles and some fish, sex is temperature-dependent (C) In birds and mammals, sex is temperature-dependent (D) In all animals, sex is genetically determined

ANSWER: (B) In reptiles and some fish, sex is temperature-dependent

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: How do sex chromosomes differ between males and females in humans?

OPTIONS: (A) Males have XX and females have XY (B) Males have XY and females have XX (C) Both have XY (D) Both have XX

ANSWER: (B) Males have XY and females have XX

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: What are voluntary actions?

OPTIONS: (A) Actions that occur without our control (B) Actions that are controlled by the spinal cord (C) Actions that we can control consciously (D) Actions that occur during sleep

ANSWER: (C) Actions that we can control consciously

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: Which part of the brain is responsible for precision of voluntary actions?

OPTIONS: (A) Cerebrum (B) Cerebellum (C) Medulla (D) Hypothalamus

ANSWER: (B) Cerebellum

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: What is the role of the medulla in the brain?

OPTIONS: (A) It controls thinking and memory (B) It controls balance and posture (C) It controls involuntary actions like heartbeat and breathing (D) It controls voluntary movements

ANSWER: (C) It controls involuntary actions like heartbeat and breathing

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Which organ in humans helps in the formation of male germ cells?

OPTIONS: (A) Prostate gland (B) Testis (C) Vas deferens (D) Penis

ANSWER: (B) Testis

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: What condition favours the formation of male germ cells?

OPTIONS: (A) High temperature (B) Low temperature (C) Room temperature (D)

Variable temperature

ANSWER: (B) Low temperature

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: What is the pathway followed by male germ cells to reach the site of fertilisation?

OPTIONS: (A) Testis → Penis → Urethra (B) Testis → Vas deferens → Urethra → Penis

(C) Testis → Urethra → Vas deferens → Penis (D) Testis → Prostate gland → Vas deferens

ANSWER: (B) Testis → Vas deferens → Urethra → Penis

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: What is the main physical feature of a male germ cell that helps it move?

OPTIONS: (A) Large size (B) Tail/Flagellum (C) Round shape (D) Tough outer wall

ANSWER: (B) Tail/Flagellum

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Name two non-surgical methods to avoid unwanted pregnancies.

OPTIONS: (A) Vasectomy and Tubectomy (B) Barrier methods and Oral pills (C) Only abstinence (D) Only condoms

ANSWER: (B) Barrier methods and Oral pills

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Name the surgical methods to prevent pregnancy.

OPTIONS: (A) Vasectomy and Tubectomy (B) Barrier methods and IUDs (C) Oral pills and Injections (D) Condoms and Diaphragms

ANSWER: (A) Vasectomy and Tubectomy

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Which of the following is a sexually transmitted disease caused by bacteria?

OPTIONS: (A) HIV/AIDS (B) Syphilis (C) Hepatitis B (D) Genital warts

ANSWER: (B) Syphilis

YEAR: 2026

CHAPTER: 1 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Which of the following is a sexually transmitted disease caused by virus?

OPTIONS: (A) Syphilis (B) Gonorrhoea (C) HIV/AIDS (D) Chlamydia

ANSWER: (C) HIV/AIDS

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CHAPTER: 2 - Chemical Reactions and Equations (Chemistry)

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YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: (i) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{NaNO}_3 + \text{AgCl}$ (ii) $\text{K}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{KCl}$.

Which option correctly describes both reactions?

OPTIONS: (A) (i) is double displacement, (ii) is displacement (B) Both are displacement reactions and precipitation reactions (C) Both are double displacement reactions and precipitation reactions (D) (i) is displacement, (ii) is double displacement

ANSWER: (C) Both are double displacement reactions and precipitation reactions

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The gases evolved on heating lead (II) nitrate crystals are:

OPTIONS: (A) NO and O_2 (B) N_2 and NO_2 (C) NO_2 and H_2 (D) NO_2 and O_2

ANSWER: (D) NO_2 and O_2

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a poor conductor of electricity?

OPTIONS: (A) Pb (B) Cu (C) Ag (D) Al

ANSWER: (A) Pb

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The natural sources of oxalic acid, lactic acid and methanoic acid respectively are:

OPTIONS: (A) tomato, curd, ant-sting (B) tomato, orange, nettle-sting (C) orange, milk, ant-sting (D) orange, sour milk, nettle-sting

ANSWER: (A) tomato, curd, ant-sting

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: When an element 'X' reacts with water, it starts floating. Identify 'X':

OPTIONS: (A) Potassium (B) Calcium (C) Sodium (D) Iron

ANSWER: (B) Calcium

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Carbon shares its valence electrons with other atoms.

Reason (R): Shared electrons help both atoms attain noble gas configuration.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Amphoteric oxides are those that:

OPTIONS: (A) React only with acids to form salts (B) React only with bases to form salts (C) React with both acids and bases to form salt and water (D) Do not react with any substance

ANSWER: (C) React with both acids and bases to form salt and water

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a basic oxide?

OPTIONS: (A) ZnO (B) Na₂O (C) CO₂ (D) Al₂O₃

ANSWER: (B) Na₂O

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an acidic oxide?

OPTIONS: (A) ZnO (B) Na₂O (C) CO₂ (D) Al₂O₃

ANSWER: (C) CO₂

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: In the reaction $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$, which substance is oxidised?

OPTIONS: (A) ZnO (B) C (C) Zn (D) CO

ANSWER: (B) C

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: In the reaction $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$, which substance is reduced?

OPTIONS: (A) ZnO (B) C (C) Zn (D) CO

ANSWER: (A) ZnO

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The balanced equation for $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$ is:

OPTIONS: (A) $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$ (B) $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$ (C) $2\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$ (D) $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow \text{PbI} + \text{KNO}_3$

ANSWER: (B) $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an example of electrolytic decomposition?

OPTIONS: (A) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (B) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ (C) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (D) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$

ANSWER: (B) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an example of decomposition by sunlight?

OPTIONS: (A) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (B) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ (C) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$ (D) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

ANSWER: (C) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In the chlor-alkali process, what is the product formed at the anode?

OPTIONS: (A) Hydrogen (B) Chlorine (C) Sodium hydroxide (D) Oxygen

ANSWER: (B) Chlorine

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In the chlor-alkali process, what is the product formed at the cathode?

OPTIONS: (A) Hydrogen (B) Chlorine (C) Sodium hydroxide (D) Oxygen

ANSWER: (A) Hydrogen

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What is the chemical formula of baking soda?

OPTIONS: (A) Na_2CO_3 (B) NaHCO_3 (C) NaOH (D) NaCl

ANSWER: (B) NaHCO_3

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What is the chemical formula of bleaching powder?

OPTIONS: (A) CaCO_3 (B) Ca(OH)_2 (C) CaOCl_2 (D) CaCl_2

ANSWER: (C) CaOCl_2

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What is the chemical formula of Plaster of Paris?

OPTIONS: (A) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (B) $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$ (C) CaSO_4 (D) CaO

ANSWER: (B) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Why are carbonate or sulphide ores converted to oxides before extraction of metals?

OPTIONS: (A) Oxides are easier to reduce than carbonates or sulphides (B) Oxides are more abundant in nature (C) Carbonates and sulphides are non-reactive (D) Oxides have higher melting points

ANSWER: (A) Oxides are easier to reduce than carbonates or sulphides

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Which reaction shows aluminium used as a reducing agent to obtain metal from its oxide?

OPTIONS: (A) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$ (B) $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ (C) $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ (D) $\text{MgO} + \text{C} \rightarrow \text{Mg} + \text{CO}$

ANSWER: (A) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: How is copper obtained from its ore Cu_2S ?

OPTIONS: (A) By heating Cu_2S in air: $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$; $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$ (B) By dissolving Cu_2S in acid (C) By electrolysis of Cu_2S (D) By reduction with carbon

ANSWER: (A) By heating Cu_2S in air: $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$; $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why are covalent compounds poor conductors of electricity?

OPTIONS: (A) They have no free electrons or ions to carry charge (B) They are insoluble in water (C) They have high melting points (D) They are non-polar

ANSWER: (A) They have no free electrons or ions to carry charge

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why does soap not form lather in hard water?

OPTIONS: (A) Soap reacts with Ca^{2+} and Mg^{2+} ions to form insoluble scum (B) Soap dissolves completely in hard water (C) Hard water has too much soap (D) Soap is ineffective in all water

ANSWER: (A) Soap reacts with Ca^{2+} and Mg^{2+} ions to form insoluble scum

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why does carbon show catenation but silicon does not?

OPTIONS: (A) C-C bond is very strong while Si-Si bond is weak (B) Silicon is larger in size (C) Carbon has 4 valence electrons while silicon has 6 (D) Silicon cannot form bonds with other atoms

ANSWER: (A) C-C bond is very strong while Si-Si bond is weak

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Oxidation of ethanol by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ produces:

OPTIONS: (A) Ethene (B) Ethanoic acid (C) Ethyl ethanoate (D) Carbon dioxide

ANSWER: (B) Ethanoic acid

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Hydrogenation of ethene produces:

OPTIONS: (A) Ethane (B) Ethyne (C) Ethanol (D) Ethene

ANSWER: (A) Ethane

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an example of a displacement reaction?

OPTIONS: (A) Combustion of natural gas (B) Reaction of a reactive metal with dilute acid (C) Electrolysis of water (D) Reaction of quick lime with water

ANSWER: (B) Reaction of a reactive metal with dilute acid

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: When dilute HCl is added to copper oxide, the solution becomes blue-green due to formation of:

OPTIONS: (A) Copper(II) oxide (B) Copper(I) chloride (C) Copper(II) chloride (D) Copper(I) oxide

ANSWER: (C) Copper(II) chloride

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: In the equation $x\text{KI} + \text{H}_2\text{O}_2 \rightarrow \text{I}_2 + y\text{KOH}$, x and y respectively are:

OPTIONS: (A) 2,2 (B) 1,2 (C) 2,1 (D) 1,1

ANSWER: (A) 2,2

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The reaction of calcium with cold water is:

OPTIONS: (A) Less violent but hydrogen catches fire (B) Very violent and hydrogen catches fire (C) Exothermic but hydrogen does not catch fire (D) Slow and endothermic

ANSWER: (C) Exothermic but hydrogen does not catch fire

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: When zinc reacts with NaOH, the product formed is:

OPTIONS: (A) Na_2O (B) ZnO (C) Na_2ZnO_2 (D) Zn(OH)_2

ANSWER: (C) Na_2ZnO_2

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): There are many compounds with carbon atoms linked to each other. Reason (R): C-C bond is very strong and stable.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Compound 'A' is used in soda-lime fire-extinguisher and decomposes on heating to form 'B'. 'B' is used to remove permanent hardness of water. Identify 'A' and 'B'.

OPTIONS: (A) A = CaCO_3 , B = CaO (B) A = CaO , B = Ca(OH)_2 (C) A = NaHCO_3 , B = Na_2CO_3 (D) A = Ca(OH)_2 , B = CaCO_3

ANSWER: (A) A = CaCO_3 , B = CaO

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The chemical formula of Plaster of Paris is:

OPTIONS: (A) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (B) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (C) CaSO_4 (D) CaO

ANSWER: (B) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

YEAR: 2026

CHAPTER: 2 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: At what temperature is gypsum heated to obtain Plaster of Paris?

OPTIONS: (A) 100°C (B) 373 K (C) 373°C (D) 100 K

ANSWER: (B) 373 K

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CHAPTER: 3 - Acids, Bases and Salts (Chemistry)

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YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following can be used as an acid-base indicator by a visually impaired student?

OPTIONS: (A) Turmeric (B) Vanilla essence (C) Methyl orange (D) Litmus

ANSWER: (B) Vanilla essence

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Tooth decay is related to pH because:

OPTIONS: (A) High pH in mouth causes tooth decay (B) Low pH due to acid formation by bacteria causes tooth decay (C) Neutral pH causes tooth decay (D) pH has no relation to tooth decay

ANSWER: (B) Low pH due to acid formation by bacteria causes tooth decay

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: How can tooth decay be prevented?

OPTIONS: (A) By eating more sugar (B) By brushing with fluoride toothpaste and avoiding sugary foods (C) By drinking more acidic drinks (D) By not brushing teeth

ANSWER: (B) By brushing with fluoride toothpaste and avoiding sugary foods

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Tamarind is a natural source of which acid?

OPTIONS: (A) Acetic acid (B) Tartaric acid (C) Oxalic acid (D) Citric acid

ANSWER: (B) Tartaric acid

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Rain water is called acid rain when its pH:

OPTIONS: (A) falls below 7.2 (B) falls below 14 (C) is above 7 (D) falls below 5.6

ANSWER: (D) falls below 5.6

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which compound is used to remove permanent hardness of water?

OPTIONS: (A) Bleaching powder (B) Washing soda (C) Baking soda (D) Caustic soda

ANSWER: (B) Washing soda

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: When stung by a honey bee, rubbing baking soda on the area gives relief because:

OPTIONS: (A) Bee sting contains an acid which is neutralized by baking soda (B) Bee sting contains a base which is neutralized by baking soda (C) Baking soda is acidic and neutralizes the sting (D) Baking soda has a cooling effect

ANSWER: (A) Bee sting contains an acid which is neutralized by baking soda

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Which type of indicator is used to predict how strong an acid or base is?

OPTIONS: (A) Universal indicator (B) Litmus paper (C) Phenolphthalein (D) Methyl orange

ANSWER: (A) Universal indicator

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Three solutions A (pH 3), B (pH 5) and C (pH 7). Which has the highest concentration of H^+ ions?

OPTIONS: (A) A (B) B (C) C (D) All have equal H^+ ions

ANSWER: (A) A

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Strong acids are those that:

OPTIONS: (A) Completely dissociate to give H^+ ions in solution (B) Partially dissociate to give H^+ ions in solution (C) Do not dissociate at all (D) Only dissociate in the presence of a base

ANSWER: (A) Completely dissociate to give H^+ ions in solution

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 4

QUESTION: Weak acids are those that:

OPTIONS: (A) Completely dissociate to give H^+ ions in solution (B) Partially dissociate to give H^+ ions in solution (C) Do not dissociate at all (D) Only dissociate in the presence of a base

ANSWER: (B) Partially dissociate to give H^+ ions in solution

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: An aqueous solution turns red litmus blue. Excess addition of which solution would reverse the change?

OPTIONS: (A) Sodium hydroxide (B) Hydrochloric acid (C) Baking soda (D) Slaked lime

ANSWER: (B) Hydrochloric acid

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: How will you protect yourself from heat while diluting a concentrated acid?

OPTIONS: (A) Add acid to water with constant stirring (B) Add water to acid with constant stirring (C) Add water to acid followed by base (D) Add base to acid with constant stirring

ANSWER: (A) Add acid to water with constant stirring

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which alloy contains a non-metal as one of its constituents?

OPTIONS: (A) Brass (B) Bronze (C) Amalgam (D) Steel

ANSWER: (D) Steel

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Pure ethanoic acid is also known as glacial acetic acid.

Reason (R): Pure ethanoic acid freezes during winter due to its low melting point of 290 K.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following are decomposition reactions? (i) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (ii) $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2$ (iii) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ (iv) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$

OPTIONS: (A) (i) and (ii) (B) (iii) and (iv) (C) (i) and (iii) (D) (ii) and (iv)

ANSWER: (C) (i) and (iii)

YEAR: 2026

CHAPTER: 3 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Carbon shows the property of catenation. Reason (R): This is due to strong C-C bonds.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

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CHAPTER: 4 - Metals and Non-Metals (Chemistry)

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YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a poor conductor of electricity?

OPTIONS: (A) Pb (B) Cu (C) Ag (D) Al

ANSWER: (A) Pb

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which metal neither reacts with cold water nor hot water but reacts with steam?

OPTIONS: (A) Sodium (B) Calcium (C) Iron (D) Magnesium

ANSWER: (C) Iron

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following metals does not react with cold water?

OPTIONS: (A) Sodium (B) Potassium (C) Calcium (D) Magnesium

ANSWER: (D) Magnesium

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which alloy contains a non-metal as one of its constituents?

OPTIONS: (A) Brass (B) Bronze (C) Amalgam (D) Steel

ANSWER: (D) Steel

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: When an element 'X' reacts with water, it starts floating. Identify 'X':

OPTIONS: (A) Potassium (B) Calcium (C) Sodium (D) Iron

ANSWER: (B) Calcium

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Compound 'A' is used in soda-lime fire-extinguisher and decomposes to form 'B'. 'B' removes permanent hardness. Identify 'A' and 'B'.

OPTIONS: (A) A = CaCO_3 , B = CaO (B) A = CaO , B = Ca(OH)_2 (C) A = NaHCO_3 , B = Na_2CO_3 (D) A = Ca(OH)_2 , B = CaCO_3

ANSWER: (A) A = CaCO_3 , B = CaO

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: White silver chloride turns grey when exposed to sunlight because:

OPTIONS: (A) It absorbs moisture from air (B) It decomposes to silver and chlorine gas (C) It reacts with oxygen to form silver oxide (D) It melts due to heat

ANSWER: (B) It decomposes to silver and chlorine gas

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Lead (II) nitrate on heating releases brown fumes because:

OPTIONS: (A) It decomposes to form nitrogen dioxide (NO_2) gas (B) It decomposes to form lead oxide (C) It oxidises to form nitrogen (D) It reduces to form lead

ANSWER: (A) It decomposes to form nitrogen dioxide (NO_2) gas

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Bags containing oil and fat are flushed with nitrogen gas to:

OPTIONS: (A) Prevent oxidation of food (B) Increase the weight of food (C) Keep food fresh by adding moisture (D) Increase the shelf life by killing bacteria

ANSWER: (A) Prevent oxidation of food

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Which method is used to convert a carbonate ore to its metal oxide?

OPTIONS: (A) Roasting (B) Calcination (C) Electrolysis (D) Reduction

ANSWER: (B) Calcination

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Which method is used to convert a sulphide ore to its metal oxide?

OPTIONS: (A) Roasting (B) Calcination (C) Electrolysis (D) Reduction

ANSWER: (A) Roasting

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The reaction of aluminium with iron (III) oxide is used for:

OPTIONS: (A) Making aluminium utensils (B) Joining cracked railway tracks (C) Producing aluminium metal (D) Making electrical wires
ANSWER: (B) Joining cracked railway tracks

YEAR: 2026
CHAPTER: 4 - Metals and Non-Metals
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 3
QUESTION: In electrolytic refining of copper, which material is used as the anode?
OPTIONS: (A) Pure copper (B) Impure copper (C) Graphite (D) Iron
ANSWER: (B) Impure copper

YEAR: 2026
CHAPTER: 4 - Metals and Non-Metals
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 3
QUESTION: In electrolytic refining of copper, which material is used as the cathode?
OPTIONS: (A) Pure copper (B) Impure copper (C) Graphite (D) Iron
ANSWER: (A) Pure copper

YEAR: 2026
CHAPTER: 4 - Metals and Non-Metals
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 4
QUESTION: What is anode mud?
OPTIONS: (A) The pure metal deposited on the cathode (B) The insoluble impurities that settle at the bottom of the anode (C) The electrolyte solution (D) The metal dissolved in the electrolyte
ANSWER: (B) The insoluble impurities that settle at the bottom of the anode

YEAR: 2026
CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why do ionic compounds have high melting and boiling points?

OPTIONS: (A) They have weak intermolecular forces (B) They have strong electrostatic forces of attraction between ions (C) They are covalent compounds (D) They are insoluble in water

ANSWER: (B) They have strong electrostatic forces of attraction between ions

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why is solder (alloy of lead and tin) used for welding electrical wires?

OPTIONS: (A) It has high melting point (B) It has low melting point (C) It is a good conductor (D) It is inexpensive

ANSWER: (B) It has low melting point

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why cannot carbon reduce the oxides of Na or Mg?

OPTIONS: (A) Na and Mg are less reactive than carbon (B) Na and Mg are more reactive than carbon (C) Carbon is not available (D) Carbon is a non-metal

ANSWER: (B) Na and Mg are more reactive than carbon

YEAR: 2026

CHAPTER: 4 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Compound 'X' reacts with aluminium to join railway tracks. Identify 'X'.

OPTIONS: (A) Iron oxide (Fe_2O_3) (B) Copper oxide (CuO) (C) Magnesium oxide (MgO) (D) Zinc oxide (ZnO)

ANSWER: (A) Iron oxide (Fe_2O_3)

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CHAPTER: 5 - Carbon and its Compounds (Chemistry)

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YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The functional group for aldehyde is:

OPTIONS: (A) -COOH (B) -CHO (C) -CO- (D) -OH

ANSWER: (B) -CHO

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Carbon shares its valence electrons with other atoms.

Reason (R): Shared electrons help both atoms attain noble gas configuration.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Hydrocarbons with general formula C_nH_{2n} represent:

OPTIONS: (A) alkanes (B) alkenes (C) alkynes (D) cyclic compounds

ANSWER: (B) alkenes

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The functional group for carboxylic acid is:

OPTIONS: (A) -OH (B) -COOH (C) -CO- (D) -CHO

ANSWER: (B) -COOH

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which set does NOT belong to the same homologous series?

OPTIONS: (A) CH_4 and C_4H_{10} (B) C_2H_6 and C_3H_8 (C) C_3H_8 and C_5H_{12} (D) C_4H_8 and C_5H_{12}

ANSWER: (D) C_4H_8 and C_5H_{12}

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an amphoteric oxide?

OPTIONS: (A) Na_2O (B) K_2O (C) CO_2 (D) Al_2O_3

ANSWER: (D) Al_2O_3

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following will NOT undergo addition reaction?

OPTIONS: (A) C_4H_8 (B) C_2H_2 (C) C_3H_8 (D) C_2H_4

ANSWER: (C) C_3H_8

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following compounds will undergo addition reaction?

OPTIONS: (A) C_4H_{10} (B) C_4H_8 (C) C_3H_8 (D) CH_4

ANSWER: (B) C_4H_8

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: A carbon compound P (C_2H_6O) is neutral to litmus and reacts with sodium to produce a gas with pop sound. Identify P.

OPTIONS: (A) Methanol (B) Ethanol (C) Ethanoic acid (D) Methanoic acid

ANSWER: (B) Ethanol

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: When ethanol is heated with conc. H_2SO_4 at 443 K, what is the role of H_2SO_4 ?

OPTIONS: (A) Oxidizing agent (B) Reducing agent (C) Dehydrating agent (D) Catalyst

ANSWER: (C) Dehydrating agent

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: What is observed when ethanol is heated with ethanoic acid in presence of an acid catalyst?

OPTIONS: (A) A sweet-smelling ester is formed (B) A foul-smelling gas is evolved (C) A precipitate is formed (D) No reaction occurs

ANSWER: (A) A sweet-smelling ester is formed

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: Why can carbon neither form C^{4+} cation nor C^{4-} anion?

OPTIONS: (A) Carbon has 4 valence electrons and requires too much energy to lose or gain 4 electrons (B) Carbon is not reactive (C) Carbon is a non-metal (D) Carbon has only 2 shells

ANSWER: (A) Carbon has 4 valence electrons and requires too much energy to lose or gain 4 electrons

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: What type of bond is formed between carbon and hydrogen?

OPTIONS: (A) Ionic bond (B) Covalent bond (C) Metallic bond (D) Hydrogen bond

ANSWER: (B) Covalent bond

YEAR: 2026

CHAPTER: 5 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: How is ethanol converted to ethanoic acid?

OPTIONS: (A) By oxidation with acidified $K_2Cr_2O_7$ (B) By reduction with H_2 (C) By dehydration with H_2SO_4 (D) By esterification with ethanol

ANSWER: (A) By oxidation with acidified $K_2Cr_2O_7$

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CHAPTER: 6 - Periodic Classification of Elements (Chemistry)

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YEAR: 2026

CHAPTER: 6 - Periodic Classification of Elements

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The functional group for aldehyde is:

OPTIONS: (A) -COOH (B) -CHO (C) -CO- (D) -OH

ANSWER: (B) -CHO

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CHAPTER: 7 - Light - Reflection and Refraction (Physics)

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YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: A convex lens of focal length 15 cm forms an image of the same size as the object. The object and image distances are:

OPTIONS: (A) -15 cm and -15 cm (B) -15 cm and +15 cm (C) -30 cm and +30 cm (D) -30 cm and -30 cm

ANSWER: (C) -30 cm and +30 cm

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: When looking at an object very close to the eyes:

OPTIONS: (A) Ciliary muscles contract and eye lens becomes thick (B) Ciliary muscles relax and eye lens becomes thick (C) Ciliary muscles contract and eye lens becomes thin (D) Ciliary muscles relax and eye lens becomes thin

ANSWER: (A) Ciliary muscles contract and eye lens becomes thick

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): White light passing through a prism gives a spectrum of seven colours. Reason (R): This is due to scattering of light, red bends minimum and violet maximum.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (C) A is true but R is false

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Which gives negative magnification greater than one?

OPTIONS: (A) Concave mirror, real image (B) Concave mirror, virtual image (C) Convex mirror, real image (D) Convex mirror, virtual image

ANSWER: (A) Concave mirror, real image

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The correct sequence of colours from top to bottom in a prism spectrum is:

OPTIONS: (A) Red, Green, Yellow, Violet, Blue (B) Red, Yellow, Green, Blue, Violet (C) Violet, Blue, Green, Yellow, Red (D) Violet, Red, Green, Yellow, Blue

ANSWER: (B) Red, Yellow, Green, Blue, Violet

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: The focal length of the eye lens changes when viewing objects at different distances because:

OPTIONS: (A) The eye lens changes its curvature by the action of ciliary muscles (B) The retina moves forward and backward (C) The pupil size changes (D) The cornea changes shape

ANSWER: (A) The eye lens changes its curvature by the action of ciliary muscles

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: A concave lens is used as a corrective lens for myopia because:

OPTIONS: (A) It diverges light rays so they focus on the retina (B) It converges light rays so they focus on the retina (C) It changes the shape of the eye lens (D) It increases the size of the image

ANSWER: (A) It diverges light rays so they focus on the retina

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: What is the expression for magnification produced by a lens?

OPTIONS: (A) $m = v/u$ (B) $m = h'/h = v/u$ (C) $m = u/v$ (D) $m = h/h' = v/u$

ANSWER: (B) $m = h'/h = v/u$

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A 4 cm tall object is placed 10 cm from a convex lens of focal length 20 cm. The size of the image is:

OPTIONS: (A) 4 cm (B) 8 cm (C) 2 cm (D) 6 cm

ANSWER: (B) 8 cm

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Scattering of light is different from reflection because:

OPTIONS: (A) Scattering sends light in all directions while reflection sends light in one direction (B) Scattering only occurs in air while reflection occurs on surfaces (C)

Scattering changes the colour of light while reflection does not (D) Scattering only works for white light

ANSWER: (A) Scattering sends light in all directions while reflection sends light in one direction

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A convex lens of focal length 20 cm forms an image when an object is placed at 40 cm. The image position is:

OPTIONS: (A) 20 cm from lens (B) 40 cm from lens (C) 60 cm from lens (D) 80 cm from lens

ANSWER: (B) 40 cm from lens

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A convex lens of focal length 20 cm forms an image when an object is placed at 40 cm. The nature of the image is:

OPTIONS: (A) Virtual and erect (B) Real and inverted (C) Virtual and inverted (D) Real and erect

ANSWER: (B) Real and inverted

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A lens combination has a convex lens of 30 cm and a concave lens of 15 cm. The equivalent focal length is:

OPTIONS: (A) +30 cm (B) +15 cm (C) -30 cm (D) -15 cm

ANSWER: (C) -30 cm

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The power of the lens combination (convex 30 cm + concave 15 cm) is:

OPTIONS: (A) +3.33 D (B) -3.33 D (C) +2.0 D (D) -2.0 D

ANSWER: (B) -3.33 D

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: The resistivity of a wire of 0.01 cm radius and 1.0 cm length with resistance 7 Ω is:

OPTIONS: (A) $2.2 \times 10^{-8} \Omega\text{m}$ (B) $2.2 \times 10^{-6} \Omega\text{m}$ (C) $2.2 \times 10^{-4} \Omega\text{m}$ (D) $2.2 \times 10^{-2} \Omega\text{m}$

ANSWER: (A) $2.2 \times 10^{-8} \Omega\text{m}$

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: An electric heater rated 220 V, 11 A when operated at 200 V consumes power:

OPTIONS: (A) 2000 W (B) 1800 W (C) 2200 W (D) 2400 W

ANSWER: (A) 2000 W

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: Which mirror always forms a virtual, erect and diminished image?

OPTIONS: (A) Plane mirror (B) Convex mirror (C) Concave mirror (D) All mirrors

ANSWER: (B) Convex mirror

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A current-carrying conductor placed in a magnetic field experiences a force because:

OPTIONS: (A) The magnetic field exerts a force on moving charges (B) The conductor creates its own magnetic field (C) The conductor gets heated (D) The conductor becomes charged

ANSWER: (A) The magnetic field exerts a force on moving charges

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An electron beam moving horizontally from back to front is deflected to the right. The magnetic field direction is:

OPTIONS: (A) Upward (B) Downward (C) Left (D) Right

ANSWER: (A) Upward

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The pattern of magnetic field due to a current-carrying wire depends on:

OPTIONS: (A) The shape of the wire (B) The material of the wire (C) The colour of the wire (D) The temperature of the wire

ANSWER: (A) The shape of the wire

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The strength of magnetic field is maximum at a point:

OPTIONS: (A) Closer to the wire (B) Farther from the wire (C) At the middle of the wire (D) At the ends of the wire

ANSWER: (A) Closer to the wire

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Tyndall effect is the:

OPTIONS: (A) Scattering of light by particles in a colloidal solution (B) Reflection of light from a smooth surface (C) Refraction of light in a prism (D) Dispersion of light in a prism

ANSWER: (A) Scattering of light by particles in a colloidal solution

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Danger signals are always red because:

OPTIONS: (A) Red light is scattered the least (B) Red light is scattered the most (C) Red light is the brightest (D) Red light is the cheapest

ANSWER: (A) Red light is scattered the least

YEAR: 2026

CHAPTER: 7 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Far-sightedness is found in which eye defects?

OPTIONS: (A) Myopia and cataract (B) Myopia and hypermetropia (C) Presbyopia and hypermetropia (D) Presbyopia and myopia

ANSWER: (C) Presbyopia and hypermetropia

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CHAPTER: 8 - Human Eye and Colourful World (Physics)

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YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: When looking at an object very close to the eyes:

OPTIONS: (A) Ciliary muscles contract and eye lens becomes thick (B) Ciliary muscles relax and eye lens becomes thick (C) Ciliary muscles contract and eye lens becomes thin (D) Ciliary muscles relax and eye lens becomes thin

ANSWER: (A) Ciliary muscles contract and eye lens becomes thick

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): White light passing through a prism gives a spectrum of seven colours. Reason (R): This is due to scattering of light, red bends minimum and violet maximum.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (C) A is true but R is false

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The vision defect arising from weakening of ciliary muscles and diminishing flexibility of eye lens is:

OPTIONS: (A) Myopia (B) Hypermetropia (C) Presbyopia (D) Astigmatism

ANSWER: (C) Presbyopia

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The pair of eye parts responsible for admitting different amounts of light is:

OPTIONS: (A) Iris and pupil (B) Ciliary muscles and pupil (C) Retina and Iris (D) Ciliary muscles and cornea

ANSWER: (A) Iris and pupil

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Far-sightedness is found in which eye defects?

OPTIONS: (A) Myopia and cataract (B) Myopia and hypermetropia (C) Presbyopia and hypermetropia (D) Presbyopia and myopia

ANSWER: (C) Presbyopia and hypermetropia

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The phenomenon responsible for rainbow formation is:

OPTIONS: (A) Refraction, dispersion and internal reflection (B) Only refraction (C) Only dispersion (D) Reflection and refraction

ANSWER: (A) Refraction, dispersion and internal reflection

YEAR: 2026

CHAPTER: 8 - Human Eye and Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: In a reflecting telescope, the image of a star seen in the eye-piece is:

OPTIONS: (A) Real and inverted (B) Virtual and erect (C) Real and erect (D) Virtual and inverted

ANSWER: (A) Real and inverted

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CHAPTER: 9 - Electricity (Physics)

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YEAR: 2026

CHAPTER: 9 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: According to Ohm's law, which graph is obtained when voltage is plotted against current?

OPTIONS: (A) Straight line passing through origin (B) Parabola (C) Hyperbola (D) Circle

ANSWER: (A) Straight line passing through origin

YEAR: 2026

CHAPTER: 9 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A wire with steeper slope in V-I graph has:

OPTIONS: (A) Lower resistance (B) Higher resistance (C) Same resistance (D) Zero resistance

ANSWER: (B) Higher resistance

YEAR: 2026

CHAPTER: 9 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The focal length of a convex lens can be found when:

OPTIONS: (A) Object is at infinity and image is at focus (B) Object is at focus and image is at infinity (C) Object is at 2F and image is at 2F (D) All of the above

ANSWER: (D) All of the above

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CHAPTER: 10 - Magnetic Effects of Electric Current (Physics)

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YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Direction of force on a current-carrying conductor in a magnetic field remains same if both magnetic field and current are reversed. Reason

(R): The direction of force can be determined using a magnetic compass.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (C) A is true but R is false

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): A compass needle kept in an external magnetic field always aligns in the North-South direction of Earth. Reason (R): The compass needle behaves like a freely suspended bar magnet.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (D) A is false but R is true

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A current-carrying conductor in a magnetic field experiences a force. This is demonstrated by:

OPTIONS: (A) Displacement of the conductor when current passes (B) Heating of the conductor (C) Change in colour of the conductor (D) Sound produced by the conductor

ANSWER: (A) Displacement of the conductor when current passes

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An electron beam moving horizontally from back to front is deflected to the right. The magnetic field direction is:

OPTIONS: (A) Upward (B) Downward (C) Left (D) Right

ANSWER: (A) Upward

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The pattern of magnetic field due to a current-carrying wire depends on:

OPTIONS: (A) The shape of the wire (B) The material of the wire (C) The colour of the wire (D) The temperature of the wire

ANSWER: (A) The shape of the wire

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The strength of magnetic field is maximum at a point:

OPTIONS: (A) Closer to the wire (B) Farther from the wire (C) At the middle of the wire (D) At the ends of the wire

ANSWER: (A) Closer to the wire

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: The rule that gives the direction of force on a current-carrying conductor in a magnetic field is:

OPTIONS: (A) Right-hand thumb rule (B) Fleming's left-hand rule (C) Fleming's right-hand rule (D) Maxwell's rule

ANSWER: (B) Fleming's left-hand rule

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: A fuse is called a safety device because:

OPTIONS: (A) It increases voltage (B) It melts and breaks the circuit when current exceeds safe limit (C) It stores charge (D) It controls power

ANSWER: (B) It melts and breaks the circuit when current exceeds safe limit

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: Electrical appliances with metallic bodies are connected with a three-pin plug while bulbs use a two-pin plug because:

OPTIONS: (A) Metallic appliances need earth connection for safety (B) Three-pin plug provides higher voltage (C) Metallic appliances consume more power (D) Three-pin plug is cheaper

ANSWER: (A) Metallic appliances need earth connection for safety

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: A burnt-out fuse should be replaced by another fuse of identical rating because:

OPTIONS: (A) A higher rating fuse may not melt during overloading (B) A lower rating fuse will melt too quickly (C) Different rating fuses have different colours (D) Identical rating fuses are cheaper

ANSWER: (A) A higher rating fuse may not melt during overloading

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: Soft iron is used in making electromagnets because:

OPTIONS: (A) It is easily magnetized and demagnetized (B) It has high melting point (C) It is a good conductor of electricity (D) It is inexpensive

ANSWER: (A) It is easily magnetized and demagnetized

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Tyndall effect is the:

OPTIONS: (A) Scattering of light by particles in a colloidal solution (B) Reflection of light from a smooth surface (C) Refraction of light in a prism (D) Dispersion of light in a prism

ANSWER: (A) Scattering of light by particles in a colloidal solution

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Danger signals are always red because:

OPTIONS: (A) Red light is scattered the least (B) Red light is scattered the most (C) Red light is the brightest (D) Red light is the cheapest

ANSWER: (A) Red light is scattered the least

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The three wires in a domestic electric circuit are:

OPTIONS: (A) Live, Neutral and Earth (B) Positive, Negative and Ground (C) Phase, Neutral and Earth (D) Active, Passive and Neutral

ANSWER: (A) Live, Neutral and Earth

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The standard potential difference supplied in Indian homes for lighting is:

OPTIONS: (A) 110 V (B) 220 V (C) 230 V (D) 240 V

ANSWER: (B) 220 V

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The role of earthing in domestic electric circuits is:

OPTIONS: (A) To prevent electric shock by providing a path for leakage current (B) To increase voltage (C) To reduce current (D) To store electricity

ANSWER: (A) To prevent electric shock by providing a path for leakage current

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The role of fuse in domestic electric circuits is:

OPTIONS: (A) To prevent damage due to overloading (B) To increase current (C) To store charge (D) To measure voltage

ANSWER: (A) To prevent damage due to overloading

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: When two bar magnets X and Y are placed near each other, and X is displaced towards Y, there will be:

OPTIONS: (A) Repulsion if like poles face each other (B) Attraction if like poles face each other (C) No force between them (D) Both attraction and repulsion

ANSWER: (A) Repulsion if like poles face each other

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A metallic conductor in a magnetic field gets displaced towards the left. If current is increased, displacement will:

OPTIONS: (A) Increase (B) Decrease (C) Remain same (D) Reverse direction

ANSWER: (A) Increase

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If a stronger horseshoe magnet is used, displacement of the conductor will:

OPTIONS: (A) Increase (B) Decrease (C) Remain same (D) Reverse direction

ANSWER: (A) Increase

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If the direction of current through a conductor is reversed, displacement will:

OPTIONS: (A) Reverse direction (B) Increase (C) Decrease (D) Remain same

ANSWER: (A) Reverse direction

YEAR: 2026

CHAPTER: 10 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: An earth wire is necessary for appliances with metallic covers because:

OPTIONS: (A) It prevents electric shock in case of leakage (B) It increases the efficiency of the appliance (C) It reduces electricity bills (D) It makes the appliance work faster

ANSWER: (A) It prevents electric shock in case of leakage

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CHAPTER: 12 - Our Environment (Biology)
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YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which group contains only non-biodegradable materials?

OPTIONS: (A) Vegetable peels, dead leaves, paper (B) Cow dung, leather bag, water
(C) Polythene bag, rubber band, ball pen (D) Paper, fruits, bones

ANSWER: (C) Polythene bag, rubber band, ball pen

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of trophic levels in a food chain is limited by:

OPTIONS: (A) Availability of water (B) Deficient food supply (C) Decrease in energy at
higher levels (D) Pollution in air

ANSWER: (C) Decrease in energy at higher levels

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): There are generally more individuals at lower trophic levels.

Reason (R): The greatest number is of the producers.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A
and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A
is false but R is true

ANSWER: (B) Both A and R are true but R is not the correct explanation of A

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Organisms of a higher trophic level feeding on several types of organisms
from lower trophic levels form the:

OPTIONS: (A) Food web (B) Ecosystem (C) Ecological pyramid (D) Food chain

ANSWER: (A) Food web

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In accumulation of toxic substances through a food chain, which organism is affected the most?

OPTIONS: (A) Small aquatic plants (B) Human beings (C) Small fish (D) Large fish

ANSWER: (B) Human beings

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which organisms belong to the same trophic level?

OPTIONS: (A) Snake and Grasshopper (B) Grass and tree (C) Lizard and elephant (D) Bacteria and grass

ANSWER: (B) Grass and tree

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Opening of stomata is regulated by:

OPTIONS: (A) Entry of water into guard cells (B) Exit of water from guard cells (C) Entry of air into stomatal pore (D) Entry of water into stomatal pore

ANSWER: (A) Entry of water into guard cells

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Highest percentage of pesticides is found in tertiary consumers. Reason (R): Chemicals accumulate progressively at each trophic level.

OPTIONS: (A) Both A and R are true and R is the correct explanation of A (B) Both A and R are true but R is not the correct explanation of A (C) A is true but R is false (D) A is false but R is true

ANSWER: (A) Both A and R are true and R is the correct explanation of A

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a non-biodegradable waste?

OPTIONS: (A) Vegetable peels (B) Paper (C) Plastic (D) Cotton cloth

ANSWER: (C) Plastic

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which gas causes depletion of the ozone layer?

OPTIONS: (A) Carbon dioxide (B) Methane (C) Chlorofluorocarbons (CFCs) (D) Sulphur dioxide

ANSWER: (C) Chlorofluorocarbons (CFCs)

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The UNEP agreement of 1987 regarding CFCs was to:

OPTIONS: (A) Ban all CFCs completely (B) Freeze CFC production at 1986 levels (C) Increase CFC production (D) Replace CFCs with other chemicals

ANSWER: (B) Freeze CFC production at 1986 levels

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The role of ozone in the atmosphere is to:

OPTIONS: (A) Absorb harmful UV radiation (B) Produce oxygen (C) Help in photosynthesis (D) Cause global warming

ANSWER: (A) Absorb harmful UV radiation

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which type of respiration provides more energy and why?

OPTIONS: (A) Aerobic respiration because glucose is completely oxidised (B) Anaerobic respiration because it does not require oxygen (C) Both provide equal energy (D) Neither provides energy

ANSWER: (A) Aerobic respiration because glucose is completely oxidised

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which type of heart has 2 chambers?

OPTIONS: (A) Fish (B) Amphibians (C) Reptiles (D) Birds

ANSWER: (A) Fish

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which organisms have a 3-chambered heart?

OPTIONS: (A) Fish (B) Amphibians and Reptiles (C) Birds and Mammals (D) Insects
ANSWER: (B) Amphibians and Reptiles

YEAR: 2026
CHAPTER: 12 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 3
QUESTION: Which type of circulation is most efficient?
OPTIONS: (A) Single circulation (B) Double circulation (C) Open circulation (D) No circulation
ANSWER: (B) Double circulation

YEAR: 2026
CHAPTER: 12 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 3
QUESTION: In xylem tissue, water is transported through:
OPTIONS: (A) Transpiration pull and cohesion of water molecules (B) Diffusion (C) Active transport (D) Osmosis
ANSWER: (A) Transpiration pull and cohesion of water molecules

YEAR: 2026
CHAPTER: 12 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 3
QUESTION: When Mendel crossed pure tall with pure short pea plants, the F₁ generation showed:
OPTIONS: (A) All tall plants (B) All short plants (C) 3 tall : 1 short (D) 1 tall : 1 short
ANSWER: (A) All tall plants

YEAR: 2026
CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: Some plants minimise which process to survive in dry conditions?

OPTIONS: (A) Photosynthesis (B) Transpiration (C) Respiration (D) Translocation

ANSWER: (B) Transpiration

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 4

QUESTION: Which part of the leaf is responsible for exchange of gases during photosynthesis?

OPTIONS: (A) Stomata (B) Xylem (C) Phloem (D) Cuticle

ANSWER: (A) Stomata

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: One advantage of asexual reproduction is:

OPTIONS: (A) Genetic variation (B) Rapid increase in population (C) Adaptability to changing environment (D) Survival in unfavourable conditions

ANSWER: (B) Rapid increase in population

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: One disadvantage of asexual reproduction is:

OPTIONS: (A) It is slow (B) Lack of genetic variation (C) It requires two parents (D) It produces fewer offspring

ANSWER: (B) Lack of genetic variation

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Asexual reproduction lacks genetic variation because:

OPTIONS: (A) Offspring are clones of the parent (B) Offspring have different genes (C) Offspring are produced by fusion of gametes (D) Offspring have new combinations of genes

ANSWER: (A) Offspring are clones of the parent

YEAR: 2026

CHAPTER: 12 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 5

QUESTION: Higher organisms resort to sexual reproduction in spite of its complexity because:

OPTIONS: (A) It produces more offspring (B) It provides genetic variation which helps in survival (C) It is faster (D) It requires less energy

ANSWER: (B) It provides genetic variation which helps in survival

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CHAPTER: 13 - Management of Natural Resources (Biology)

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YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which group contains only non-biodegradable materials?

OPTIONS: (A) Vegetable peels, dead leaves, paper (B) Cow dung, leather bag, water (C) Polythene bag, rubber band, ball pen (D) Paper, fruits, bones

ANSWER: (C) Polythene bag, rubber band, ball pen

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the best way to dispose of kitchen waste?

OPTIONS: (A) Burning (B) Composting (C) Dumping (D) Burying

ANSWER: (B) Composting

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: What is the best way to dispose of plastic bottles?

OPTIONS: (A) Burning (B) Recycling (C) Dumping (D) Composting

ANSWER: (B) Recycling

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: One harmful effect of using plastic bags is:

OPTIONS: (A) They decompose easily (B) They are non-biodegradable and cause pollution (C) They are expensive (D) They are biodegradable

ANSWER: (B) They are non-biodegradable and cause pollution

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: A better alternative to plastic bags is:

OPTIONS: (A) Paper bags (B) Metal bags (C) Glass bags (D) Rubber bags

ANSWER: (A) Paper bags

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Pesticides and chemical fertilizers from paddy fields reach our bodies through:

OPTIONS: (A) Food chain (B) Direct consumption (C) Air (D) Skin contact

ANSWER: (A) Food chain

YEAR: 2026

CHAPTER: 13 - Management of Natural Resources

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: The phenomenon by which chemicals accumulate progressively at each trophic level is called:

OPTIONS: (A) Bioaccumulation (B) Biomagnification (C) Biodegradation (D)

Biosynthesis

ANSWER: (B) Biomagnification